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Crystal Defects & Solid Solutions

Magnitude of the Burgers Vector in α-Fe (BCC) and Al (FCC)

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> *6. Calculate the Burgers vector magnitude for* α-Fe *(BCC) and* Al *(FCC).* *(Solution:* α-Fe : $\frac{\sqrt{3}a}{2}$*;* Al : $\frac{a}{\sqrt{2}}$*)*
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> **Solution:** α-Fe : $\frac{\sqrt{3}a}{2}$; Al : $\frac{a}{\sqrt{2}}$

PHASE 1: HYPOTHESIS & PARAMETERS

Lattice Models:

1. **Ferritic Iron (**α-Fe**):** Body-Centred Cubic (BCC) lattice, lattice parameter a.
2. **Pure Aluminium (**Al**):** Face-Centred Cubic (FCC) lattice, lattice parameter a.

Fundamental Hypothesis:

- In any metallic crystal lattice, stable dislocations are **unit (or perfect) dislocations** whose Burgers vector $\vec{b}$ connects two identical, equivalent lattice positions along the **crystallographic direction of maximum packing** in order to minimise the elastic energy of the dislocation ($E \propto |\vec{b}|^2$) [Session 4 Slides 35–36].

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

Citing the theory of dislocations and slip systems [Session 4 Slides 35–37]:

- **In the BCC lattice:**
- The direction of maximum atomic packing is the main cube diagonal $\langle 111 \rangle$.
- The smallest lattice translation connecting two lattice nodes is the one going from a vertex $(0, 0, 0)$ to the body centre $(1/2, 1/2, 1/2)$:

$$\vec{b}_{\text{BCC}} = \frac{a}{2}[1\,1\,1]$$

- **In the FCC lattice:**
- The direction of maximum packing is the face diagonal $\langle 110 \rangle$.

- The smallest lattice translation joining two identical lattice positions goes from a vertex $(0,0,0)$ to the face centre $(1/2,1/2,0)$:

$$\vec{b}_{\mathrm{FCC}} = \frac{a}{2}[1\,1\,0]$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Magnitude in α-Fe (BCC):

- Burgers vector in Cartesian components:

$$\vec{b} = \left(\frac{a}{2}, \frac{a}{2}, \frac{a}{2}\right)$$

- Magnitude (Euclidean norm):

$$|\vec{b}| = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2} = \sqrt{3\left(\frac{a^2}{4}\right)}$$

$$|\vec{b}|_{\alpha\text{-Fe}} = \frac{\mathbf{a\sqrt{3}}}{\mathbf{2}} = \frac{\mathbf{\sqrt{3}a}}{\mathbf{2}}$$

(Note that since in BCC $a\sqrt{3} = 4R$, the magnitude is exactly $|\vec{b}| = 2R$, which corresponds to the atomic diameter in contact).

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2. Magnitude in Al (FCC):

- Burgers vector in Cartesian components:

$$\vec{b} = \left(\frac{a}{2}, \frac{a}{2}, 0\right)$$

- Magnitude (Euclidean norm):

$$|\vec{b}| = \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2 + 0^2} = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \sqrt{\frac{2a^2}{4}} = \frac{a\sqrt{2}}{2}$$

Rationalising by dividing numerator and denominator by $\sqrt{2}$:

$$|\vec{b}|_{\text{Al}} = \frac{\mathbf{a}}{\sqrt{\mathbf{2}}}$$

(Note that in FCC $a\sqrt{2} = 4R$, so $|\vec{b}| = \frac{4R}{2} = 2R$, which also coincides exactly with the atomic diameter in contact).

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Universality of the Atomic Diameter:** In both metallic crystal structures, the magnitude of the Burgers vector of the elementary perfect dislocation is identical to the atomic diameter:

$$|\vec{b}| = 2R$$

- This shows mathematically why plastic slip occurs step by step along the dense atomic rows: each time the dislocation line sweeps a plane, it displaces the upper block of the crystal by one elementary atomic distance $|\vec{b}| = 2R$ relative to the lower one.